

Amir Karimi

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Experience

- Aug 2022–present **Doctoral Researcher**, KTH Royal Institute of Technology, Stockholm, Sweden
Representation learning, causal and object-centric inductive biases, and training dynamics of deep transformers.
- Apr–Jun 2026 **Visiting Researcher**, Technical University of Munich & Helmholtz AI, Munich, Germany
Progressive depth growth for efficient pretraining of self-supervised vision transformers.
Supervised by Stefan Bauer.
- Dec 2023–Mar 2024 **Visiting Researcher**, Technical University of Munich & Helmholtz AI, Munich, Germany
Evaluation of Bayesian causal discovery (ICML 2024).
Supervised by Stefan Bauer.
- Feb 2020–May 2022 **Research Assistant**, Data Analytics Lab, University of Tehran, Tehran, Iran
Master's thesis research on the lifespans of Bitcoin and Ethereum addresses (ICCKE 2022).
Supervised by Behnam Bahrak.
- 2017–2022 **Teaching Assistant**, University of Tehran, Tehran, Iran
Courses: Deep Learning, Discrete Mathematics, Algorithm Design, Data Structures, Engineering Probability and Statistics, Database Design, Programming Languages and Compilers.
- Aug 2017–Jul 2020 **Teacher**, Allameh Helli High School (Exceptional Talents), Iran
Courses: C++ Programming, Web Development.
- Nov 2019–Mar 2020 **Data Engineer**, Tapsell, Iran
Developed an IP geolocation model and contributed to advertising recommendation and fraud detection methods.
- Jul–Aug 2017 **Software Engineer Intern**, Rahnema College, Iran
Developed a location-based social networking application for Android.

Education

- Aug 2022–Feb 2027 (expected) **PhD in Computer Science**, KTH Royal Institute of Technology, Stockholm, Sweden
Thesis topic: *Representation Structure, Evaluation, and Depth Dynamics in Deep Models*.
Licentiate of Engineering, 2025.
Licentiate thesis: *Bayesian Causal Discovery and Object-Centric Representations: Challenges and Insights in Structured Learning*.
Supervised by Karl Henrik Johansson and Stefan Bauer.
- Sep 2019–May 2022 **MSc in Computer Engineering (Artificial Intelligence and Robotics)**, University of Tehran, Tehran, Iran
Thesis: *Longevity Analysis of Cryptocurrency Addresses*.
Supervised by Behnam Bahrak.
- Sep 2015–Jul 2019 **BSc in Software Engineering**, University of Tehran, Tehran, Iran
Thesis: *Classification of Electrocardiogram Signals using Machine Learning and Neural Networks*.
Supervised by Siamak Mohammadi.

Selected Publications

- Zhou, Y., Golob, J. L., **Karimi Mamaghan, A. M.**, Bauer, S.[†], & Schwab, P.[†] [ViroGym: Realistic Large-Scale Benchmarks for Evaluating Viral Proteins](#). NeurIPS 2026 (Evaluations and Datasets Track, spotlight).
- **Karimi Mamaghan, A. M.**, Tao, Y., Niazi, M. U. B., Bauer, S., & Johansson, K. H. From Prediction to Structure: Causal Discovery for 30-Day Readmission Prediction. IEEE BHI 2026.
- Kapl, F., **Karimi Mamaghan, A. M.**, Seitzer, M., Johansson, K. H., Marr, C., Bauer, S., & Dittadi, A. [Are Object-Centric Representations Better at Compositional Generalization?](#) ICML 2026.
- Siji, A.^{*}, **Karimi Mamaghan, A. M.**^{*}, Kapl, F., Höppe, T., Angelis, E., Dittadi, A., Brenner, M., Heinzinger, M., Johansson, K. H., Maile, K., von Oswald, J., & Bauer, S. [From Words to Amino Acids: Does the Curse of Depth Persist?](#) ICLR 2026 LMRL Workshop; arXiv:2602.21750.
- **Karimi Mamaghan, A. M.**, Papa, S., Johansson, K. H., Bauer, S., & Dittadi, A. [Exploring the Effectiveness of Object-Centric Representations in Visual Question Answering: Comparative Insights with Foundation Models](#). ICLR 2025.
- **Karimi Mamaghan, A. M.**, Tigas, P., Johansson, K. H., Gal, Y., Annadani, Y., & Bauer, S. [Challenges and Considerations in the Evaluation of Bayesian Causal Discovery](#). ICML 2024.
- **Karimi Mamaghan, A. M.**, Dittadi, A., Bauer, S., Johansson, K. H., & Quinzan, F. [Diffusion-Based Causal Representation Learning](#). Entropy, 2024.
- Wang, Z.^{*}, **Karimi Mamaghan, A. M.**^{*}, De Santi, R., Hsieh, Y.-P., Bauer, S., & Johansson, K. H. Diffusion Adaptation Under Test-Time Distribution Shifts via Distributional Optimization. *Under review at AISTATS 2027*.
- **Karimi Mamaghan, A. M.**^{*}, Kapl, F.^{*}, Dittadi, A., Johansson, K. H., Maile, K., von Oswald, J., & Bauer, S. Growing to See: Efficient Pretraining of Self-Supervised Vision Transformers. *Under review at ICLR 2027*.
- **Karimi Mamaghan, A. M.**, Kapl, F., Dittadi, A., Johansson, K. H., & Bauer, S. Object-Centric Learning for Part-Whole Hierarchies: Promises, Limitations, and Open Challenges. *Under review at a NeurIPS 2026 workshop*.
- Siji, A., **Karimi Mamaghan, A. M.**, Kapl, F., Höppe, T., Dittadi, A., Johansson, K. H., Heinzinger, M., & Bauer, S. Exploring Compute Efficiency and Task Transfer in Depth-Grown Protein Language Models. *Under review at a NeurIPS 2026 workshop*.
- Gruber, C., **Karimi Mamaghan, A. M.**, Pauline, V., Höppe, T., & Bauer, S. Identification and Reconstruction of Human Endogenous Retroviral (HERV) Sequences. *Manuscript in preparation*, 2026.
- Gruber, C., Krus, A. L., **Karimi Mamaghan, A. M.**, Pauline, V., Zhou, Y., & Bauer, S. High-Affinity Protein Binder Design with Protein Foundation Models. *Manuscript in preparation*, 2026.

^{*}Equal contribution. [†]Shared last authors.

Academic Service

Reviewer ICLR, ICML, NeurIPS, and CLeaR

Awards and Scholarships

- 2026 ICML Gold Reviewer Award
- 2026 KTH Travel Grant from the Styffe Foundation for NeurIPS 2026
- 2026 WASP Research Stint Abroad funding (research visit, Apr–Jun 2026)
- 2025 ELLIS PhD Student (nominated by Karl Henrik Johansson and Stefan Bauer)
- 2025 Erasmus+ Scholarship (research visit to Munich, Jul 2025)
- 2024 HIDA Visiting Researcher Grant (research visit, Dec 2023–Mar 2024)
- 2022 Full doctoral funding, WASP Graduate School (2022–2027)
- 2015–2018 Faculty of Engineering Prize for three consecutive years; ranked 1st among BSc Software Engineering students, University of Tehran
- 2015 Ranked 91st of about 200,000 (top 0.05%) in Iran’s university entrance exam

Skills

Technical	Python, PyTorch; previous experience: R, C/C++, Kotlin, Java, Django, Spring Boot, Kubernetes
Tools	Git, Slurm (HPC), LaTeX
Languages	Persian, English